AI Email Assistant for Laptop Enquiries

Assignment write-up: n8n + Pinecone (vector database) + Google Gemini

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1. Summary

I built an automated assistant that **reads customer emails, finds the right laptops in a vector database, and replies on its own** within a few minutes. A customer can ask things like *"What is the price of the Lenovo Legion Y520-15IKBN?"* or *"Show me your new HP laptops"* and receive an accurate, friendly reply in the same email thread.

Every email is labelled in Gmail (**Processed / Ignored / Failed**) and recorded in a Google Sheet log, so the team can always see what the system did.

2. How it works

#	Step	What happens
1	Receive email	n8n checks the company Gmail inbox every minute for new unread emails, one email at a time.
2	Safety check	Automated senders (no-reply, alerts, newsletters) are skipped before using any AI.
3	Understand	Gemini reads the email and turns it into structured data: type of email, laptop names, what the customer wants, and filters (brand, budget, RAM, year).
4	Build the query	Code turns that data into a Pinecone search with filters, e.g. brand = Lenovo, or brand = HP and year = 2026.
5	Search	Pinecone finds the most similar laptops by meaning; code then keeps only exact model matches.
6	Prepare facts	Code counts versions, works out the price range and formats each laptop as a ready-made line.
7	Reply and record	Gemini writes a friendly reply around those lines; Gmail sends it in the same thread, labels it, and the Google Sheet logs it.

3. Tools used

Tool	Role	Why I chose it
n8n (self-hosted, Docker)	Runs the whole workflow	Free, visual, no execution limits; data stays on my machine
Pinecone	Vector database (1,249 laptops)	Finds laptops by meaning, not exact words; built-in embedding model; free tier
Google Gemini (Flash)	Understands emails, writes replies	Fast and low cost; a backup model is used if the main one is busy
Gmail + Google Sheets	Email in/out, labels, log	Secure OAuth access; the log is easy for anyone to read

4. Data preparation

- **Source:** Kaggle laptop price dataset (1,303 rows).

- **Cleaning:** removed 28 duplicate rows, merged 26 same-laptop listings, fixed inconsistent brand spellings (e.g. "Thinkpad" and "ThinkPad"), cleaned broken product names and fixed the file encoding. Result: **1,249 laptops, no missing values.**
- **Prices:** converted from euros to US dollars at 1 EUR = 1.1176 USD (13 Oct 2026).
- **Launch_Year:** added a simulated year (2016–2026) for every laptop, so questions about "new" or "latest" laptops can be answered. These years are for demo purposes only.
- **Loading:** each laptop is stored in Pinecone as a short descriptive sentence (for meaning-based search) plus its details as metadata (for exact filters).

5. Edge cases handled

Situation	What the system does
Laptop not found	Polite reply with 3 different close alternatives
Many versions / several laptops	Gives the version count and price range; answers each laptop asked about
General question (budget, brand, newest)	Recommends up to 5 matching laptops
Unrelated email or automated sender	No reply (prevents spam replies and reply loops); labelled Ignored and logged
AI or database failure	Retries first; then sends a polite holding reply, emails an alert to the team, labels Failed and logs the error

6. Testing

I tested every path with real emails while the workflow was live, and checked every price and spec in the replies against the dataset:

- Single laptop with 19 versions, specifications request, two laptops in one email, new HP laptops, laptop not found: all replies correct.
- Unrelated email and automated sender: correctly ignored with no reply.
- Failure: when Gemini was overloaded, the customer received a holding reply and I received an alert, as designed.

7. Problems I found and fixed

- **Wrong price in an early reply:** the AI mixed up two nearly identical versions. I moved all facts and calculations into code, so the AI only writes the wording. All later replies matched the data exactly.
- **AI service overload:** added automatic retries and a backup model, so busy periods no longer cause failed replies.
- **Confusing similar versions:** added screen details to each line, and made sure "not found" suggestions show different models.

8. Possible improvements

- Host n8n on a cloud server so it runs 24/7, not only when my computer is on.
- Use real launch dates and live prices from a product database.
- Add a human approval step for large orders, and a simple dashboard built on the log.
